

BioMechAI — Deep Learning Model Architecture Manual

Comprehensive Engineering Guide to All Systems Combining to Form Our Action Recognition Champion

ARCHITECTURE
BLUEPRINT

1. Overview: The 7 Integrated Architectural Systems

Our action recognition engine does not rely on a single algorithm in isolation. Instead, it is an end-to-end spatiotemporal machine learning pipeline combining on-device computer vision, geometric kinematic formatting, and 3D deep convolutional neural networks.

Pipeline Flow: Raw Video -> ML Kit 3D Skeletons -> 48-Frame Temporal Sampler -> 17 Connected Limb Heatmaps -> SlowOnly 3D ResNet-50 Backbone -> 3D Global Average Pooling -> I3D Classification Head -> 7 Softmax Classes

2. Detailed Explanation of Combining Components

Component 1: Google ML Kit On-Device 3D Pose Detection

EDGE VISION

What it is: A lightweight neural detector running directly on the mobile phone CPU at 30 frames per second.

How we use it: It takes the camera feed and outputs normalized 3-dimensional coordinates (X, Y, Z) and likelihood scores across 33 anatomical landmarks. By extracting coordinates locally, we discard raw video pixels immediately, preserving complete user privacy and shielding the AI from background clutter and lighting changes.

Component 2: Temporal Golden Window Sampler (48 Frames, Stride 2)

TIME NORMALIZATION

What it is: An automated temporal resampling filter operating over our 2.5-second movement buffer (75 raw frames at 30 FPS).

How we use it: Different people perform repetitions at slightly different cadences. The sampler extracts exactly 48 uniform frames with a temporal stride of 2. This standardizes all inputs so that whether an exercise takes 2.1 seconds or 2.7 seconds, the neural network always receives an identically formatted spatiotemporal volume.

Component 3: Connected 3D Spatiotemporal Limb Heatmaps

INPUT TENSOR MODALITY

What it is: A continuous volumetric representation that connects pairs of joint coordinates into cylindrical limb segments with a Gaussian radius of $\sigma = 0.6$.

How we use it: Older approaches (like PoseC3D v4) used joint dot heatmaps, which failed on squats because when an athlete squats, their hips, knees, and ankles overlap in the sagittal camera plane (causing joint dots to merge and disappear). Our model renders 17 anatomical limb pairs (thighs, shins, forearms, torso line). Even if a joint overlaps, the connecting limb remains visible, boosting squat recognition by over 160% relative accuracy.

Component 4: The SlowOnly Pathway

EFFICIENCY ARCHITECTURE

What it is: A streamlined single-stream variant of Facebook AI Research's SlowFast video network architecture.

How we use it: In raw RGB video, a dual-pathway network is required because the AI must separate moving objects from noisy backgrounds using a high-frequency Fast pathway. But because our input consists of clean skeleton limb heatmaps on a pure black background, background noise is already eliminated. The expensive Fast pathway is redundant. The SlowOnly stream samples 48 frames at a steady pace, reducing computational complexity from 65+ GFLOPs down to 18 GFLOPs while maintaining high accuracy.

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ResNet-50 Backbone, Pooling, Classification Head, Pretraining, and Verification

PART 2: DEEP LEARNING ENGINE

Component 5: 3D ResNet-50 Convolutional Backbone (ResNet3dSlowOnly)

FEATURE EXTRACTOR

What it is: A 50-layer deep convolutional neural network that uses 3D convolutions (3x3x3) sliding across Space (56x56) and Time (48 frames) simultaneously.

How we use it: It contains residual skip connections that add inputs directly to outputs: $Output = F(Input) + Input$. This solves the vanishing gradient problem, allowing gradients to flow freely across all 50 layers during training. The backbone has 3 distinct stages with block distribution (4, 6, 3):

- **Stage 1 (4 Blocks):** Uses 2D spatial convolutions to detect static body orientation (distinguishing standing vs. prone exercises).
- **Stage 2 (6 Blocks):** Inflates into 3D spatiotemporal convolutions to track joint angle changes (knee and elbow bending).
- **Stage 3 (3 Blocks):** Analyzes whole-body cadence and repetition speed across time (distinguishing jumping vs. slow strength holds).

Component 6: 3D Global Average Pooling

DIMENSIONALITY REDUCTION

What it is: A spatiotemporal pooling layer that averages features across the entire temporal and spatial dimensions (48 frames x 7 height x 7 width -> 1x1x1).

How we use it: It condenses millions of intermediate activations into a compact, fixed-size **512-dimensional biomechanical feature vector** representing the essence of the exercise repetition.

Component 7: 13D Linear Classification Head and Softmax

DECISION OUTPUT

What it is: A fully connected projection layer with a 0.6 Dropout regularization rate followed by Softmax activation.

How we use it: It maps the 512-dimensional vector into 7 discrete logits corresponding to our target classes: [Bicep Curl, High Knees, Jumping Jack, Lunge, Plank, Push-Up, Squat]. Softmax normalizes the outputs into calibrated probabilities summing to 1.0 (e.g., Squat: 97.58%).

Component 8: Transfer Learning via FineGYM Athletic Pretrained Weights

WEIGHT INITIALIZATION

What it is: Initialization using OpenMMLab's athletic weights pretrained on FineGYM gymnastic routines (126.8 MB).

How we use it: Rather than training 31.7 million weights from scratch, the model starts with pre-learned representations of human limb dynamics. Fine-tuning these weights on our audited 3,159 clean exercise clips enabled convergence in just 10 to 18 epochs on Kaggle GPU.

3. Technical Specifications Summary Table

ARCHITECTURAL PARAMETER	CONFIGURED VALUE	SYSTEM FUNCTION
Input Tensor Dimensions	[1, 17, 48, 56, 56]	Batch size 1, 17 limb channels, 48 frames, 56x56 spatial resolution
Network Depth	50 Layers	Deep hierarchical spatiotemporal feature extraction
Spatiotemporal Inflation	(0, 1, 1)	Stage 1 spatial; Stages 2 and 3 3D spatiotemporal
Output Feature Vector	512 Dimensions	Condensed biomechanical representation
Total Model Parameters	31.7 Million Weights	Checkpoint size: 8.45 MB
Inference Latency	120 ms to 160 ms	Real-time responsiveness via FastAPI cloud endpoint

ARCHITECTURAL PARAMETER	CONFIGURED VALUE	SYSTEM FUNCTION
Top-1 Accuracy (v7 Unique)	63.54%	Evaluated on strictly unique held-out test clips

4. How to Explain the Full System to the Panel (Speech Script)

OFFICIAL PANEL PRESENTATION SCRIPT:

"Respected panel, our exercise recognition model is an integrated spatiotemporal pipeline combining five core innovations: First, Google ML Kit extracts 33 body coordinates locally, eliminating background noise. Second, we render these into 17 connected 3D limb heatmaps using Gaussian cylinders ($\sigma = 0.6$), which solves sagittal self-occlusion during squats. Third, we adopt the SlowOnly pathway, sampling 48 frames at a steady pace to cut compute by over 60 percent without losing temporal detail. Fourth, a 3D ResNet-50 backbone with selective inflation processes space and time simultaneously across 50 layers. Finally, initializing with FineGYM athletic weights and fine-tuning on our audited 3,159 clean clips allowed our champion model to reach 63.54 percent Top-1 accuracy with 61.39 percent Macro Recall."